

BILIM VA MALAKALARNI BAHOLASH AGENTLIGI DARAJASIDA

Bilimingga ishon va muvaffaqiyatga erish!

UMUMTA'LIM VA MAXSUS FANLARDAN BILISH DARAJASINI ANIQLASHNING MILLIY
SERTIFIKAT IMTIHONI ISHTIROKCHILARI UCHUN

MATEMATIKA
FANIDAN
TEST TOPSHIRIQLARI KITOB

Ishtirokchining familiyasi, ismi va otasining ismi

Imzo

Testni tuzuvchi: **TypoThesis jamoasi**

Telegram kanal: **@Milliymath_sertifikat**

Ushbu test varianti 45 ta (1-45) topshiriqdan iborat.

Test topshiriqlari RASCH MODEL asosida baholanadi.

Kitobning yopiq va ochiq turdagi test topshiriqlari mavjud.

– yopiq turdagi test topshiriqlarida javob variantlari (A, B, C va D) orasidan aynan bitta javobni tanlang va javoblar varaqasidagi topshiriq raqamiga mos qatorda belgilang.

– ochiq turdagi test topshiriqlarining javobini javoblar varaqasidagi topshiriq raqamiga mos qatorda aniq, tushunarli va eng sodda ko'inishda yozing.

– moslashtirishni talab qiluvchi yopiq test topshiriqlari uchun umumiy oltita (A – F) javob variantlari berilgan, uchta topshiriqqa (33-35 — test topshiriqlariga) ushbu javoblar orasidan mos ravishda bittadan javob tanlang va javoblar varaqasiga belgilang.

MILLIY SERTIFIKAT NAMUNAVIY SAVOLLAR TO'PLAMI!
2.03.2026. 1-smena.

1. Olti xonali $\overline{8629ab}$ soni 12 ga qoldiqsiz bo'linsa, $a \cdot b$ eng katta qiymati toping.

- A) 48
- B) 56
- C) 72
- D) 54

2. Hisoblang:

$$\frac{45\frac{10}{63} - 44\frac{25}{84}}{\left(2\frac{1}{3} - 1\frac{1}{9}\right) : 4 - \frac{3}{4}} : \frac{31}{32}.$$

- A) $-\frac{1}{2}$
- B) -2
- C) $\frac{1}{2}$
- D) 3

3. Bir sonning 10% iga, ikkinchi sonning $\frac{1}{6}$ qismini qo'shganimiz, birinchi sonning yarmidan ikkinchi sonning yarmini ayirganimizga teng. Birinchisini ikkinchisiga nisbatini toping.

- A) $\frac{3}{5}$
- B) $\frac{4}{3}$
- C) $\frac{5}{3}$
- D) $\frac{3}{4}$

4. O'quvchi 2 ta kitobni 205000 so'mga oldi. Agar birinchi kitob narxi 15% ga pasayib ikkinchisini 20% ko'tarilsa kitob narxlari teng bo'ladi. Birinchi kitob ikkinchisidan qancha qimmat?

- A) 30000
- B) 36000
- C) 33000
- D) 35000

5. Hisoblang:

$$(-1^2)^3 + (-1^3)^4 + \dots + (-1^{99})^{100}.$$

- A) -1
- B) 0
- C) 1
- D) 2

6. Hisoblang:

$$\sqrt[3]{26 + 15\sqrt{3}} + \sqrt[3]{26 - 15\sqrt{3}}.$$

- A) 1
- B) 2
- C) 3
- D) 4

7. Sonlarni o'sish tartibida joylashtiring:

$$a = \sqrt{8} + \sqrt{2}, \quad b = \sqrt{6} + 2, \quad c = 2\sqrt{3} + \sqrt{2}.$$

- A) $a < b < c$
- B) $a < c < b$
- C) $b < a < c$
- D) $c < a < b$

8. Arifmetik progressiyaning n ta hadi o'rta arifmetigi $2n$ ga teng bo'lsa, $S_{10} - S_9$ ni toping.

- A) 40
- B) 20
- C) 38
- D) 36

9. Geometrik progressiyaning hadlari uchun $b_1 + b_2 + b_3 = 70$ va $b_1 \cdot b_2 \cdot b_3 = 8000$ bo'lsa, S_5 ni toping.

- A) 300 va 70
- B) 310 va 77,5
- C) 310
- D) 70

10. Soddashtiring:

$$\frac{2y + x - \frac{4x^2 - y^2}{x}}{y^3 + 2xy^2 - 3x^2y}.$$

- A) $\frac{1}{xy}$
- B) $\frac{1}{y}$
- C) $\frac{1}{x}$
- D) $\frac{1}{xy^2}$

11. Soddashtiring:

$$\frac{\sqrt{\frac{1}{2} \left(\frac{x^2 - 2}{2x} \right)^2 + 1}}{\frac{x^2 + 2}{2x}}.$$

- A) $\sqrt{2}$
- B) $\frac{1}{2}$
- C) $\frac{1}{\sqrt{2}}$
- D) 2

12. $\alpha \in \left(\frac{3\pi}{2}; 2\pi\right)$ uchun $\sin \alpha = -\frac{4}{5}$ bo'lsa, $\cos \alpha$ ni toping.

- A) $\frac{3}{4}$
- B) $-\frac{3}{5}$
- C) $\frac{4}{5}$
- D) $\frac{3}{5}$

13. Tengsizlikni nechta butun son qanoatlantiradi:

$$\left(\frac{1}{2}\right)^{x^2+3x} - \frac{2^{-x}}{8} > 0 .$$

- A) 2 ta
- B) 1 ta
- C) 3 ta
- D) 4 ta

14. Nechta natural son tengsizlikni yechimi bo'la olmaydi:

$$x(x-3)^2 > 0 .$$

- A) 1 ta
- B) 2 ta
- C) 3 ta
- D) 4 ta

15. Tenglamaning haqiqiy ildizlari nechta:

$$(x-6)^4 + (x-8)^4 = 16 .$$

- A) 1 ta
- B) 2 ta
- C) Yechimga ega emas
- D) 3 ta

16. Tengsizlikni nechta butun yechimi bor:

$$\log_{0,5} \log_2 \log_{x-1} 9 > 0 .$$

- A) 4 ta
- B) 6 ta
- C) 5 ta
- D) 7 ta

17. Tenglamaning $\left[-\frac{\pi}{2}; \pi\right]$ oralig'idagi yechimlari sonini toping:

$$\sqrt{3} \sin^2 x + \sin^2 2x = \frac{\sqrt{3}}{2} - \frac{1}{2} - \sqrt{3} \cos 2x .$$

- A) 1 ta
- B) 2 ta
- C) 4 ta
- D) 3 ta

18. Tengsizlikni nechta butun son qanoatlantiradi:

$$\frac{x^2 - 1}{x^2 + 1} + x^2 - 2x + 1 \leq 0 .$$

- A) 1 ta
- B) 2 ta
- C) 4 ta
- D) 3 ta

19. Tenglamaning nechta haqiqiy ildizi bor:

$$\frac{4}{(x+6)(x-1)} - \frac{5}{(x+2)(x+3)} = \frac{1}{4} .$$

- A) 2 ta
- B) 1 ta
- C) 4 ta
- D) 3 ta

20. $|a| > 1$, $-1 < b < 1$, $-1 \leq c < 1$ sonlari uchun $f(x, y, z) = \frac{x^2 + 1}{x^2 - 1} + \frac{2y^2 - 1}{2y^2 + 1} + \frac{4z^2 - 1}{4z^2 + 1}$ va $(x, y, z) = \left(\frac{\sqrt{a^2 - 1}}{a - 1}; \sqrt{\frac{b^2 - 1}{2(b - 1)^2}}; \frac{\sqrt{c^2 + 1}}{2(c - 1)}\right)$ bo'lsa, $f(x, y, z)$ ni soddalashtiring.

- A) $a + b + c$
- B) 1
- C) -1
- D) $\frac{a + b + c}{2}$

21. Agar $f(x) = \ln x + \ln 2$ va $g(x) = 2e^x$ bo'lsa, $g(f(3))$ ni toping.

- A) 15
- B) 9
- C) 10
- D) 12

22. $f(x) = \frac{1}{x^2 + 4}$ funksiyaga o'tkazilgan urinma tenglamasi absissa o'qiga parallel bo'ladi. Urinish nuqtasining koordinatalari yig'indisini toping.

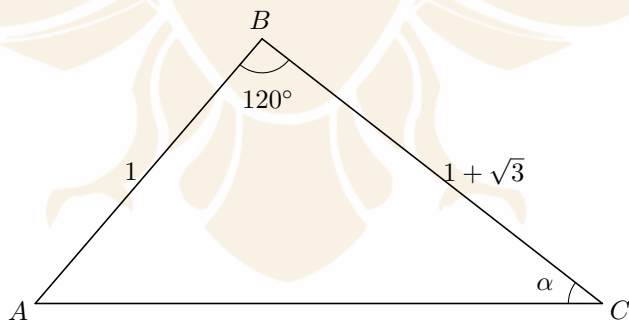
- A) 0
- B) $\frac{1}{2}$
- C) 1
- D) $\frac{1}{4}$

23. Hisoblang:

$$\int_{-1}^1 \frac{x^4}{x^2 + 1} dx .$$

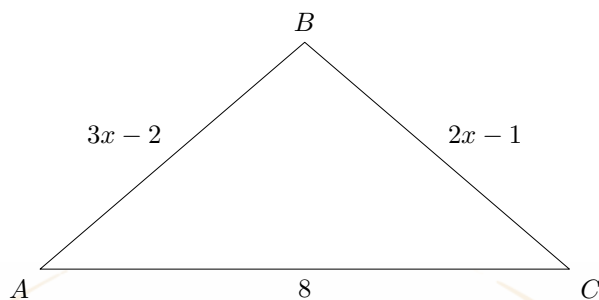
- A) $\frac{\pi}{2} - \frac{4}{3}$
- B) $\frac{\pi}{2} - \frac{3}{4}$
- C) $\frac{\pi}{2} - \frac{1}{3}$
- D) $\frac{\pi}{2} - \frac{2}{3}$

24. ABC uchburchak uchun $AB = 1$, $BC = 1 + \sqrt{3}$ va $\angle ABC = 120^\circ$ bo'lsa, $\angle BCA$ ni toping.

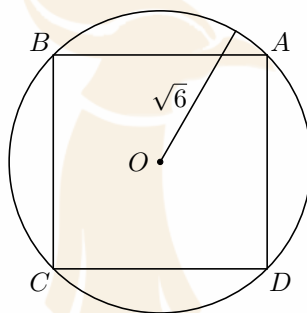


- A) 30°
- B) 60°
- C) 15°
- D) 45°

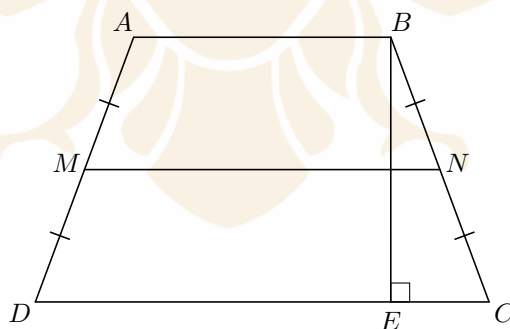
25. Tomonlari turli butun sonlardan iborat uchburchakning eng katta tomoni 8 ga teng. Yon tomonlari esa $3x - 2$ va $2x - 1$ ga teng. Perimetrini eng katta qiymatini toping.



- A) 18
B) 20
C) 21
D) 22
26. Quyidagi chizmaga asosan kvadrat yuzini toping.

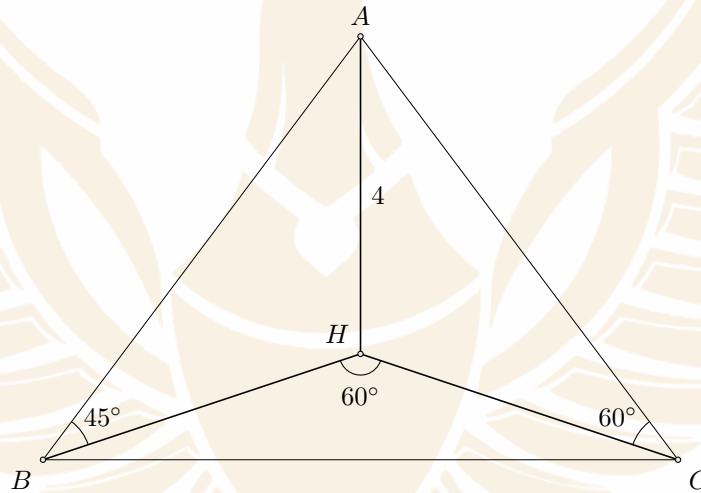


- A) 6
B) 12
C) $6\sqrt{2}$
D) $12\sqrt{2}$
27. ABCD trapetsiyada $MN = 48$ va BE balandlik uni $5 : 1$ nisbatda bo'ladi. Katta asosni toping.



- A) 54
B) 58
C) 62
D) 64
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28. n burchakli qavariq ko'pburchakning dioganallar soni 25 tadan kam emas, 30 tadan ko'p emas. n ni toping.
- A) 9
B) 10
C) 11
D) 12
29. ABC uchburchakning koordinatalari $A(2; -3; 1)$, $B(2; 1; 3)$ va $C(2; 3; 5)$ ga teng. B uchidan AC tomonga BD mediana o'tkazildi. $\angle CDB$ ning kosinusini toping.
- A) $\frac{2}{\sqrt{13}}$
B) $-\frac{3}{\sqrt{13}}$
C) $\frac{3}{\sqrt{13}}$
D) $-\frac{2}{\sqrt{13}}$
30. $ABCH$ piramidaning AH qirrasi asos tekisligiga perpendikulyar va 4 ga teng. Chizmaga asosan piramida hajmini toping.



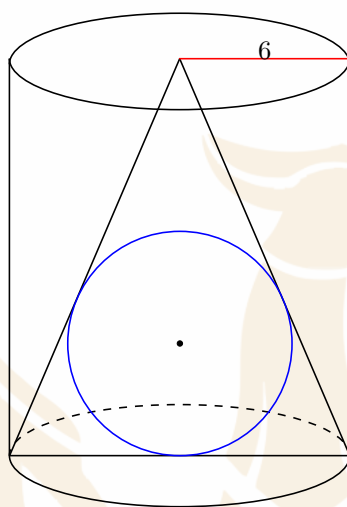
- A) 5
B) $\frac{16}{3}$
C) 6
D) $\frac{17}{3}$
31. Bir to'plamning elementlariga 1 ta qo'shilgandagi va 1 ta element olingandagi qism to'plamlari soning farqi 12 ga teng bo'lsa, dastlabki to'plamning qism to'plamlari soni toping.
- A) 64
B) 32
C) 16
D) 8

32. 1000 dan kichkina natural sonlar ichidan 11 ga bo'linadiganlari tanlab olindi. Ular ichida 2 ga va 3 ga bo'linmaydiganlari nechta?

- A) 90
- B) 45
- C) 30
- D) 15

Topshiriqlar (33–35) va javob variant (A–F) larni o'zaro moslashtiring.

Asosi 6 cm ga teng bo'lgan silindrga konus ichki chizilgan, konusga esa shar ichki chizilgan. Konusning asosi va uchi silindr asoslari markazlarida yotadi. Agar konus yon sirti 60π cm² ga teng bo'lsa,



Javoblar.

- A) 196π
- B) 168π
- C) 72π
- D) 96π
- E) 36π
- F) 192π

33. Silindr va konus to'la sirtlarining ayirmasini toping.

34. Silindr va konus hajmlari ayirmasini toping.

35. Shar hajmini toping.

36. Tenglamani yeching:

$$\sqrt{x} + 2x + 2 = (x + 3)\sqrt[4]{x}.$$

a) Tenglama nechta haqiqiy ildizga ega?

Javob: _____

b) Tenglamaning haqiqiy ildizlari yig'indisini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

37. Tenglamani yeching:

$$\sin^3 x \cdot (1 + \operatorname{ctg} x) + \cos^3 x (1 + \operatorname{tg} x) = 2\sqrt{\sin x \cdot \cos x}.$$

a) Tenglamaning eng kichik musbat ildizini toping.

Javob: _____

b) Tenglama $\left[-\frac{5\pi}{2}; \frac{5\pi}{2}\right]$ oraliqda nechta ildizga ega?

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

38. $f(x) = \frac{4x^2}{4x^4 + 1}$ funksiya berilgan.

a) Funksiyaning eng katta qiymatini toping.

Javob: _____

b) Funksiyaning qiymatlar sohasiga nechta butun son tegishli?

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

39. $f(x) = \arctg 2^{x-1013}$ funksiya berilgan.

a) $f(0) + f(1) + \dots + f(2026)$ ni hisoblang.

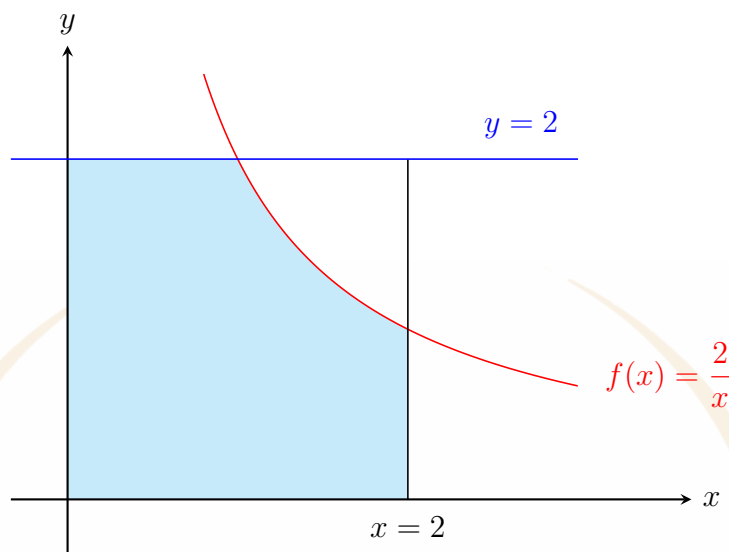
Javob: _____

b) $f'(1013)$ ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

40. Chizmada $f(x) = \frac{2}{x}$ funksiya va $y_1 = 2$, $x_1 = 2$ to'g'ri chiziqlar tasvirlangan.



a) Bo'yalgan soha yuzini toping.

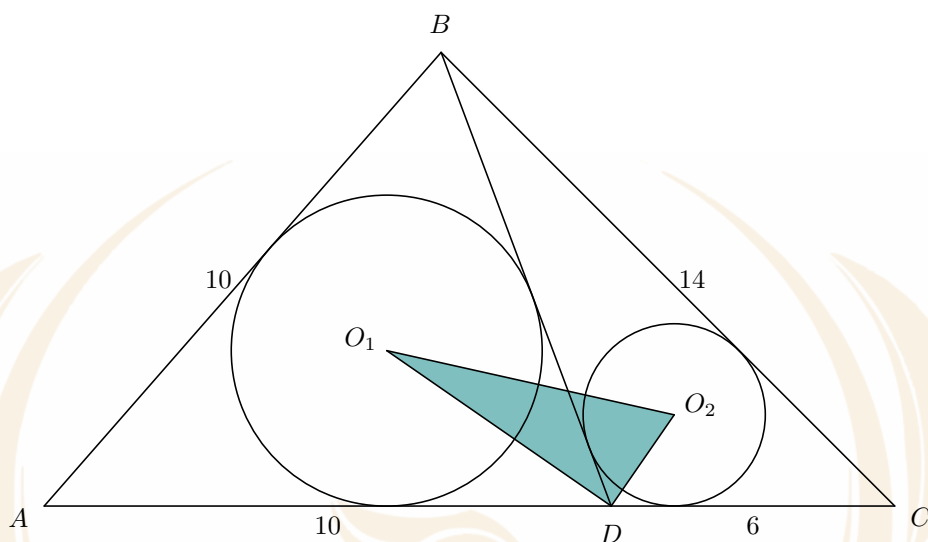
Javob: _____

b) Bo'yalgan soha yuzini Ox o'qi atrofida 360° ga aylantirishdan hosil bo'lgan jism hajmini toping. ($\pi \approx 3$ deb oling)

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

41. ABC uchburchakdan AC asosiga BD to'g'ri chiziq o'tkazilgan va ABD uchburchakga O_1 markazli, BDC uchburchakga O_2 markazli ichki aylanalar chizilgan. Agar $AB = 10$, $AD = 10$, $BC = 14$ va $DC = 6$ bo'lsa,



a) BD ni toping.

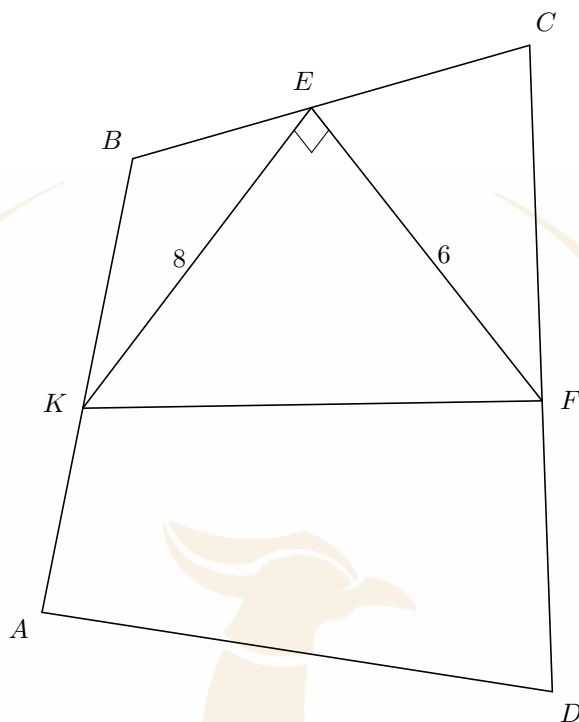
Javob: _____

b) O_1O_2D uchburchak yuzini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqarajasiga ko'chirib yozing.

42. $ABCD$ qavariq to'rtburchakning AB , BC va CD tomonlaridan mos ravishda K , E , F nuqtalar olindi. Bunda $\frac{AK}{KB} = \frac{BE}{EC} = \frac{CF}{FD} = \frac{3}{4}$, $KE = 8$, $EF = 6$ va $KE \perp EF$ bo'lsa,



a) KEF uchburchak yuzini toping.

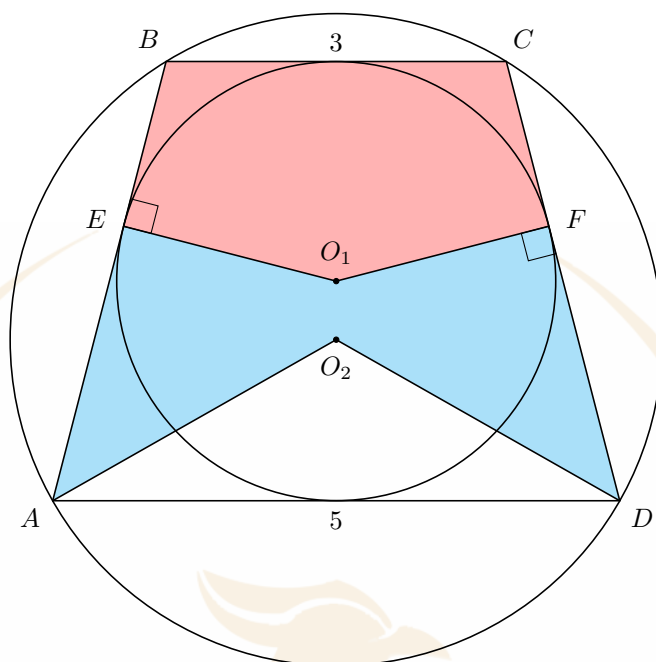
Javob: _____

b) $ABCD$ to'rtburchak yuzini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiiga ko'chirib yozing.

43. $ABCD$ trapetsiya, asoslari $AD = 5$ va $BC = 3$. Markazlari O_1 va O_2 bo'lgan aylanalar ichki va tashqi chizilgan. O_1 markazli aylana AB va CD yon tomonlarga E va F nuqtalarda urinadi.



a) $BCFO_1E$ beshburchak yuzini toping.

Javob: _____

b) AEO_1FDO_2 oltiburchak yuzini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ko'chirib yozing.

44. Sharga yon qirralari teng, asosi teng yonli trapetsiyadan iborat piramida ichki chizilgan. Trapetsiyaning yon tomonlari 3 ga, asoslari 5 va 8 ga teng. Piramida yon qirrasi $\frac{14\sqrt{3}}{3}$ ga teng.

a) Piramida balandligini toping.

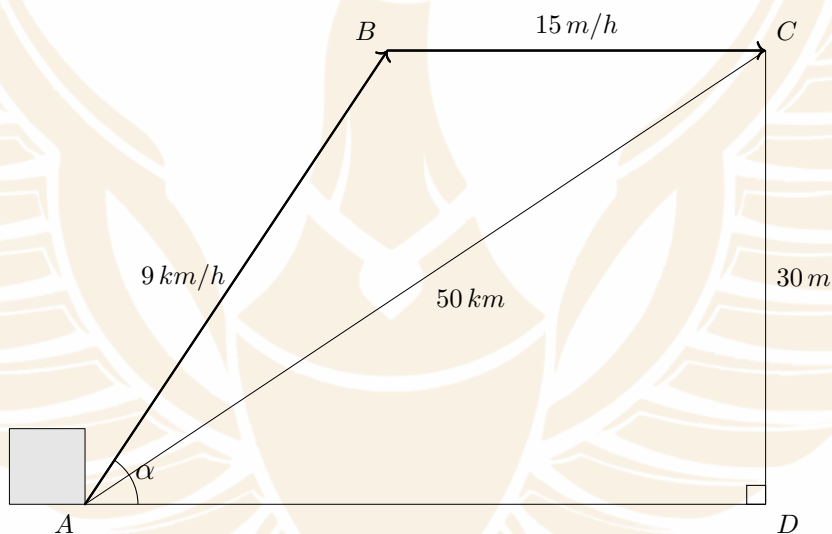
Javob: _____

b) Shar radiusini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

45. Quyida AB qiyalik orqali jism C nuqtaga qarab tortilmoqda. AB qiyalikda 9 km/h , BC tekislikda esa 15 km/h tezlik bilan tortilmoqda. $AC = 50 \text{ m}$, $DC = 30 \text{ m}$ $\angle BAD = \alpha$ bo'lsin. A nuqtadagi jismni eng qisqa vaqt ichida C nuqtaga tortib borildi. Shu shartga asosan,



a) α ni toping.

Javob: _____

b) Eng qisqa vaqtni toping.(h)

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.